MILITARY SPECIFICATION

CONTROL, INVERTER CHANGEOVER, TYPE A-2

1. SCOPE

1.1 This specification covers one type of automatic inverter changeover control, designated Type A-2.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following specifications, bulletin, and drawing, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications:

Military:

MIL-P-14	Plastic Materials, Molding, and Plastic-Parts, Molded; Thermosetting
MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories, and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)
MIL-P-6064	Packaging of Lightweight Aircraft Accessories

Air Force-Navy Aeronautical:

AN-M-13	Marking, Labeling, and Tagging, Aircraft Aeronautical Equipment, Accessories, Parts, Materials, and Supplies
AN-S-126	Screw-Threads; Standard Aircraft

2.1.2 Specification Bulletin:

Air Force-Navy Aeronautical:

143	Specifications and Standards; Use of

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2.1.3 Drawing:

U. S. Air Force:

45B7367 Control - Inverter Change Over Type A-1 Automatic

(Copies of this specification, Military, Air Force-Navy Aeronautical, and U. S. Air Force specifications, bulletins, and drawings may be obtained upon application to the Commanding General, Air Material Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Component parts.- The control shall consist of a mounting bracket, holding coil, heater switch, terminal parts, relay, and cover.

3.2 Material:

3.2.1 Specifications and standards.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143, except as provided in the following paragraph.

3.2.1.1 Standard parts.- AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.2 Protective treatment.- When materials are used in the construction of the control that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Design and construction.- The control shall be of compact design and of sufficiently rugged construction to withstand the mechanical shocks and stresses incident to their use on aircraft.

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3.3.1 The control shall be designed to permit disassembly and reassembly for the purpose of repair or replacement of parts without special tools.

3.3.2 The control shall conform substantially to Drawing No. 45B7367.

3.3.3 The control shall be constructed to insure proper operation when mounted in any position, and the complete mechanism shall be totally enclosed for mechanical and sand and dust protection.

3.3.4 Mounting bracket.- The control mounting bracket shall be an integral part of the control housing or shall be securely attached thereto in a manner to prevent any movement between the control and mounting bracket in service use.

3.3.5 Interior mechanism.- The interior mechanism of the control shall be secured so that when the cover is removed, the control shall be capable of operating satisfactorily without adjustment.

3.3.6 Phenolic.- Phenolic or any other material used in the construction of the control shall be suitably treated or protected against moisture absorption.

3.3.7 Holding coil:

3.3.7.1 Winding.- The relay holding coils shall be properly insulated and taped.

3.3.7.2 Heads.- The relay holding coil heads, if used, shall be of phenolic material conforming to Specification MIL-P-14, or equivalent.

3.3.8 Contact base.- The contact base shall be of phenolic material conforming to Specification MIL-P-14, or equivalent.

3.3.9 Contact alignment.- The movable contacts of the relay and heater switch shall be self-aligning with the stationary contacts and shall be positive in operation.

3.3.10 Spacing and creepage distances.- The minimum spacing between current-carrying parts and any other portions of the control, other than suitable insulating material, and also between current-carrying parts of opposite polarity, shall be 1/16 inch.

3.4 Performance.- The control shall be capable of satisfactory operation under the following conditions:

(1) At ambient temperatures ranging from -55 to +55°C.
(2) A pressure altitude range from sea level to 50,000 feet.
(3) Sand and dust as encountered in desert areas.
(4) Exposure to salt-laden atmosphere.
(5) Vibration incident to service use.

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3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Rating.- The rating of the relay and heater switch contacts shall not be less than 3 amp at 29V dc.

3.7 Heater switch.- The heater switch shall not exceed a maximum time delay of 5 seconds, shall not be less than 2 seconds on closing contacts, and shall open circuit within a maximum time of 10 seconds.

3.8 Terminal posts:

3.8.1 All leads of the control assembly shall be brought out on one end of the control to terminal posts and insulated from the mounting bracket. The leads shall not be exposed to possible damage in service.

3.8.2 The terminal posts shall be threaded with No. 8-32 NC-2 threads and shall be provided with the proper nuts and lock washers or suitable binding head screws equipped with lockwashers for attaching the cable terminals. The terminals shall not be exposed to possible damage in service and shall have a minimum thread allowance of 1/4 inch.

3.8.3 The terminals shall be securely fastened to the control so that the terminals cannot be twisted, pushed in, or pulled out in normal service use.

3.9 Terminal protection:

3.9.1 The portions of the terminals projecting on the outside of the control shall be separated from each other by suitable insulating barriers to provide protection against crossing-over of the external wiring.

3.9.2 A removable protection cover of suitable insulating material shall be provided to cover the terminals and barriers after the external connections have been made. The cover shall enclose the bottom and one side of the terminals, and shall be designed to be reversible in order to provide access to either side of the terminals, as desired.

3.10 Terminal markings.- The terminals shall be suitably marked as shown on Drawing No. 45B7367. These markings shall be permanent and easily readable.

3.11 Protective cover:

3.11.1 The control shall be provided with a suitable protecting cover.

3.11.2 Suitable means shall be provided for securely fastening the cover to the control base, and for permitting removal of the cover when necessary.

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3.11.3 The cover shall fit the control base snugly to prevent the entrance of corrosive atmosphere or sand or dust, and shall be capable of meeting the test specified in Section 4.

3.12 Wiring:

3.12.1 The internal wiring of the control shall be with wire capable of withstanding the insulation tests and shall be of a suitable gage to withstand the tests specified in Section 4.

3.12.2 The wires shall be arranged and secured to prevent the cable from interfering with the proper operation of any of the component relays or heater switch.

3.13 Screw threads.- Screw threads shall be in accordance with Specification AN-S-126.

3.14 Weight.- The weight of the control shall not be more than 1.1 pounds.

3.15 Finish.- All exposed steel parts shall be of corrosion-resistant steel or protected by a plating of corrosion-resistant metal.

3.16 Nameplate.- A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the control:

Control, Inverter Changeover
115V AC, 400 Cycle and 29V DC
Type A-2
Specification MIL-C-6521
Mfr's. Part No._____
Mfr's. Serial No._____
Order No._____
Stock No._____
(Mfr's. Name or Trademark)
U. S. Property

3.16.1 Manufacturer's part number.- The manufacturer's part number marked in the space provided on the nameplate shall be in accordance with Specification AN-M-13.

3.16.2 Manufacturer's serial number.- The manufacturer's serial number shall be entered in the space provided on the nameplate. The serial number shall not be repeated and shall be a concise identifying number.

3.16.3 Order number.- Where both the purchase order number, such as "AF 33(038) 50-4710" and the contract number, such as "AF 33(038)-1565", are given on an order or contract, the number placed on the nameplate in the place designated "Order Number" shall be the number corresponding to the Air Force's purchase order number, for example: "50-4710". In the absence of the Air Force's purchase order number, and the contract number is given, the letters "AF" and the last digits of the contract number shall be placed on the nameplate, for example: "AF-1565".

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3.16.4 Stock number.- The stock number shall be as specified by the Procuring Agency. The stock number shall include the name or standard abbreviations of the name of the service having jurisdiction of the equipment.

3.17 Workmanship:

3.17.1 General.- The control including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, accuracy of dimensions, and freedom from burrs and sharp edges.

3.17.2 Cleaning.- The control shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of controls shall be classified as follows:

4.1.1 Qualification tests.- Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

4.1.2 Inspection tests.- Inspection tests are those tests accomplished on controls submitted for acceptance under contract.

4.2 Test conditions:

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3 Qualification test:

4.3.1 Prior qualification.- Unless otherwise specified by the Procuring Agency, controls which have not previously passed a qualification test, or which have passed the qualification test and have been modified in any manner shall satisfactorily pass a qualification test prior to the acceptance of any control.

4.3.2 Qualification test samples.- Qualification test samples shall consist of 3 controls. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

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4.3.3 The qualification test shall consist of the following tests and those specified under Inspection tests:

4.3.3.1 Operating voltage:

4.3.3.1.1 The direct-current relay, provided with single-pole normally open, single-pole normally closed, and one single-pole double throw contact arrangement, shall be subjected to tests to determine that the normally open contacts close and the normally closed contacts open instantaneously with a direct-current potential of 18V, or less, applied to the appropriate terminals of the assembly with the heater switch contacts closed and to determine that the relay contacts remain closed and open, respectively, on 7V.

4.3.3.1.2 The 400 cycle ac relay, provided with a single pole, single-throw or double-throw contact arrangement, shall be subjected to tests to determine that the normally open contacts close with an alternating current potential of 102.5 ± 2.5V applied to the appropriate terminals of the assembly. These contacts shall open at 92.5 ± 2.5V.

4.3.3.1.3 The heater switch shall be tested to determine that the normally open contacts shall be capable of closing with a minimum time delay of 2 seconds and a maximum time delay of 5 seconds when a direct-current potential of 22 to 29V, inclusive, is applied to the heater coil. The contacts shall open circuit within a maximum time period of 10 seconds.

4.3.3.2 Current consumption:

4.3.3.2.1 Direct-current relay.- The direct-current relay shall be tested to determine that the holding coil consumes no more than 0.3 amp when operated on 29V dc.

4.3.3.2.2 Alternating-current relay.- The alternating-current relay shall be tested to determine that the holding coil consumes no more than 0.15 amp when operated on 115V ac.

4.3.3.2.3 Heater switch coil.- The heater switch coil shall be tested to determine that the holding coil consumes no more than one amp when operated on 29V dc.

4.3.3.3 Vibration resistance.- The control shall be subjected to vibration tests. Vibration shall consist of simple harmonic motion having an amplitude not to exceed 0.03 inch (maximum total excursion of 0.06 inch) and a frequency varied uniformly between the approximate limits of 10 to 55 cps. The entire cycle of frequencies shall be accomplished in approximately one minute. The control shall be mounted in the position most likely to cause malfunctioning and vibrated continuously for a period of 3 hours. If mounting in more than one position is considered likely to cause malfunctioning, the control shall be vibrated continuously for 3 hours in each likely position.

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A test lamp or other suitable indicating device shall be connected across the applicable contacts. The control shall be tested to determine that the vibration does not close the normally open contacts with 18V dc across the direct-current relay and does not open the alternating-current relay normally closed contacts with 105V ac across the coil.

4.3.3.3.1 During the vibration test, the heater switch shall be tested to determine that the contacts close with 22 to 29V, inclusive, across the coil at the specified time. The heater switch contacts shall stay open during this test with no voltage across the coil.

4.3.3.4 Acceleration resistance.- The control shall be subjected to acceleration forces of 10g. The time required to attain the 10g acceleration shall not be more than 2 seconds. The control shall be mounted in the position most likely to cause malfunctioning, and shall be accelerated for a minimum of one minute without causing the control normally open contacts to close. A test lamp or a suitable indicating device shall be connected across the contact terminals while the control is undergoing this test, to determine that the acceleration does not result in closing or opening of the applicable contacts.

4.3.3.5 Low temperature.- The control shall be subjected to -54 ± 2°C for a minimum period of 16 hours. At the end of this period and with the control at the specified temperature, the control shall be subjected to the applicable tests specified in the operating voltage test. The control shall then be tested intermittently for operating voltage characteristics until it attains a temperature of 20 to 30°C. The voltage characteristics during this test shall conform to the applicable test specified in the operating voltage test.

4.3.3.6 High temperature.- The control shall be subjected to a temperature of +71 ± 2°C for a minimum of 16 hours. At the completion of this period and with the control at the specified temperature, the control shall be subjected to the applicable test specified in the operating voltage test. The control shall then be tested intermittently for operating voltage characteristics until it attains a temperature of 20 to 30°C. The voltage characterisitcs during this test shall be in accordance with the applicable tests specified in the operating voltage test.

4.3.3.7 Continuous operation.- The control shall be subjected to tests at rated capacity for a continuous 5-hour period with a direct-current potential of 29V, and an alternating-current potential of 115V applied to the applicable terminals of the control. At the completion of this test, the operating voltage characteristics of the control shall conform to the operating voltage tests and to the insulation test specified in the Individual tests. This test shall be conducted at an ambient temperature of +71 ± 2°C.

4.3.3.8 Life test.- The control shall be mounted either in a vertical or horizontal position and subjected to a life test of 500 operations as follows: A suitable automatic test rig shall be employed which shall connect 29V dc between Terminals G and "sw", and 115V, 400 cycles between

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Terminals AC and G. After a maximum of 15 seconds, the 115V, 400-cycle circuit shall be opened and after a maximum of 25 seconds the 29V dc circuit shall be opened. After a maximum of 20 seconds "Off" period, the cycle shall be repeated. During this test, a 3 amp, 28V dc resistive load shall be connected between the main inverter and Terminal G and between the spare inverter and Terminal G. Following the 500 test cycles, the control contacts shall be tested to determine that no welding of the contacts has taken place, that the contacts provide positive electrical connections, and the heater switch functions properly.

4.3.3.9 Environmental.- The following tests shall be in accordance with the specified procedures of Specification MIL-E-5272 and as specified herein:

4.3.3.9.1 Humidity.- Procedure I. During this test, a potential of 150V rms at commercial frequency, alternating current shall be applied between the terminals and case. Immediately after this test, the control shall be subjected to the applicable operating voltage test specified herein, and then shall be subjected to 500 continuous cycles in accordance with the Life test.

4.3.3.9.2 Sand and dust.- Procedure II.

4.3.3.9.2.1 After subjection to the Sand test, the control shall be subjected to the following tests in the sequence specified, after immediately attaining a room temperature of 20 to 30°C.

(1) Operating voltage in accordance with the Operating voltage test specified herein.
(2) Five hundred continuous cycles in accordance with the Life test specified herein.

4.3.3.9.3 Fungus.- Procedure I. No fungus growth which will affect subsequent operation shall result from this test.

4.3.3.9.4 Salt spray.- Procedure I for a 50-hour period. At the completion of this test, the control shall be subjected to the Operating voltage test and the Life test specified herein.

4.3.3.9.5 Altitude.- Procedure II for a period of 4 hours. No failure of the control shall result from this test.

4.4 Inspection tests.

4.4.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

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4.4.2 Individual tests.- Each control shall be subjected to the following tests:

4.4.2.1 Examination of product.- Each control shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2.2 Insulation test.- The control shall be subjected to the following voltages between the appropriate terminals and other exposed metal parts:

4.4.2.2.1 Five hundred volts rms at 60 cycles frequency for 60 seconds between the exposed metal parts and all terminals connected to the 29V dc circuits.

4.4.2.2.2 One thousand two hundred and fifty volts rms at 60 cycles frequency for 60 seconds between the exposed metal parts and the terminals connected to the 115V, 400-cycle circuit.

4.4.3 Sampling tests.- One control shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests:

4.4.3.1 Lot.- A lot shall consist of controls manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.4.3.2 When tests are specified on a quantity of controls that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, additional controls of the lot represented shall be tested immediately to determine the extent of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the control.

4.4.3.3 Weight.- The control shall be weighed and shall not exceed the weight specified in Section 3.

4.4.3.4 The control shall be subjected to the following tests specified in the Qualification test:

(1) Operating voltage
(2) Current consumption
(3) Vibration resistance
(4) Life test

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification and/or drawings, to determine compliance with the specification and/or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

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5. PREPARATION FOR DELIVERY

5.1 General.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by, or direct shipments to the Government.

5.2 The control shall be packaged and packed for shipment in accordance with Group V of Specification MIL-P-6064, in a Type II Class I container.

5.3 Marking.- Marking shall be in accordance with Specification MIL-P-6064 and as specified herein.

5.3.1 Interior package.- Each interior package shall be marked with the following information:

Control, Inverter Changeover, Automatic
115V AC, 400 Cycle and 29V DC
Type A-2
Specification MIL-C-6521
Mfr's. Part No._____
Mfr's. Serial No._____
Order No._____
Stock No._____
Date of Manufacture_____
Quantity_____
(Mfr's. Name or Trademark)
Name of Contractor (if not the same as manufacturer)

IF IN STORAGE AFTER (DATE) (1), THE CONTROL SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

(1) This date shall be 18 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

5.3.2 Precautionary marking.- The following precautionary marking shall appear on two opposite sides of each interior package, wherever practicable, depending on the size of the carton and shall also appear on the exterior shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

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6. NOTES

6.1 Use.- The Type A-2 control covered by this specification is intended for use in aircraft as a means of automatically controlling the operation of aircraft inverters.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.3 Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Air Materiel Command and arrange to have the products that they propose to offer to the Air Force tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio, Attention: MCREXE.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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